

Sri Lanka Institute of Information Technology

Data warehousing and Business Intelligence

Assignment 1



MAYADUNNA S M M T S – IT23584068

Year 03 Semester 01 - WE 01.01

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1.Introduction

This report documents the design and implementation of a complete Data Warehouse (DW) and Business Intelligence (BI) solution for Assignment 1 of the IT3021 module. The system was built using **Microsoft SQL Server Integration Services (SSIS)** and **SQL Server Management Studio (SSMS)**.

The selected dataset is Sri Lanka Multi-Branch Pharmacy Sales OLTP dataset containing **50,000** retail pharmacy transactions across 7 branches and 6 provinces over an **18-month period (January 2022 to June 2023)**. The system demonstrates a full data warehousing pipeline from raw transactional data through staging ETL processing, dimensional modelling, and analytical reporting.

This report is structured across six tasks as defined in the assignment brief:

- Task 1: Dataset selection and justification
- Task 2: Preparation of multiple data sources
- Task 3: DW & BI solution architecture design
- Task 4: Data warehouse dimensional model design and SQL implementation
- Task 5: ETL development using SSIS packages
- Task 6: Accumulating fact table update ETL

Task 1:Dataset Selection

1.1 Dataset Overview

The dataset used for this project consists of **50,000 transactional pharmacy sales records** collected over an **18-month period from January 2022 to June 2023**. It represents the operations of a **multi-branch retail pharmacy chain in Sri Lanka**, capturing detailed real-world business activities such as medicine sales, supplier interactions, customer demographics, and payment methods.

Each record in the dataset corresponds to a **single medicine sale line item within a transaction**, making it a highly granular dataset suitable for analytical processing

Business Scenario

The dataset simulates the day-to-day operations of a pharmacy chain, including:

- Multiple branches distributed across different provinces in Sri Lanka
- Sales of both **prescription and over-the-counter (OTC)** medicines
- Supplier distribution and procurement activities
- Customer demographic behavior and purchasing patterns
- Pricing structures, discounts, and revenue generation

- Batch-level tracking and medicine expiry management

The dataset is structured as **strict OLTP (Online Transaction Processing) dataset**, meaning it is optimized for transactional operations rather than analysis. This makes it highly suitable for transformation into a **Data Warehouse (DW)**.

Time Coverage

- **Start Date:** 01 January 2022
- **End Date:** 30 June 2023
- **Duration:** 18 months

This extended time period allows for meaningful trend analysis and seasonal comparisons.

Data Volume

- **Total Records:** 50,000 transaction records
- **Branches:** 7 pharmacy branches
- **Entities:** Multiple suppliers, medicine categories, and customer segments

The dataset is suitable for this assignment because it allows comprehensive analysis of various business aspects. It can be used to track sales trends over months and years, compare performance across different branches, and analyze revenue at the category level. Additionally, it enables evaluation of a supplier performance, examination of prescription versus over-the-counter sales, and understanding of customer demographics. The dataset also supports monitoring of product expiry and batch information, as well as analysis of payment method distribution, providing valuable insights for informed decision-making.

1.2 Dataset Columns

The table below describes all 27 columns in the original OLTP dataset:

Column Name	Data Type	Description
Transaction_ID	INT	Unique identifier for each sales transaction (natural key)
Transaction_Date	DATE	Date the transaction occurred
Branch_ID	INT	Identifier of the pharmacy branch
Branch_Name	VARCHAR	Full name of the pharmacy branch
District	VARCHAR	District in which the branch is located
Province	VARCHAR	Province in which the branch is located

Payment_Method	VARCHAR	Mode of payment: Card, Cash, or Insurance
Customer_Gender	VARCHAR	Gender of the customer (Male/Female)
Customer_Age	INT	Age of the customer in years
Customer_City	VARCHAR	City from which the customer comes
Medicine_ID	INT	Unique identifier for the medicine
Medicine_Name	VARCHAR	Full branded name of the medicine
Generic_Name	VARCHAR	Active pharmaceutical ingredient name
Medicine_Category	VARCHAR	Therapeutic category (e.g., Antibiotic, Vitamin)
Dosage_Form	VARCHAR	Physical form: Tablet, Capsule, Syrup, Cream, etc.
Strength_mg	DECIMAL	Dosage strength in milligrams
Pack_Size	INT	Number of units per pack
Supplier_Name	VARCHAR	Name of the pharmaceutical supplier
Prescription_Required	VARCHAR	Whether a prescription is needed (Yes/No)
Batch_No	VARCHAR	Batch number of the medicine stock
Expiry_Date	DATE	Expiry date of the medicine batch
Quantity	INT	Number of packs sold in the transaction
Unit_Price_LKR	DECIMAL	Price per pack in Sri Lankan Rupees
Discount_Rate	DECIMAL	Discount fraction applied to the transaction
Line_Total_LKR	DECIMAL	Net revenue = UnitPrice × Qty × (1 – DiscountRate)
Year	INT	Year of transaction (derived)
Month	INT	Month of transaction (derived)

1.3 Source ER diagram

This ER model represents a pharmacy sales system centered on **transactions**, which link customers, medicines, branches, suppliers, and payments. Each transaction is tied to single customer and branch, while medicines and suppliers can participate in multiple transactions. Payment methods record how sales are completed. The design ensures data integrity by clearly defining relationships and dependencies among all entities.

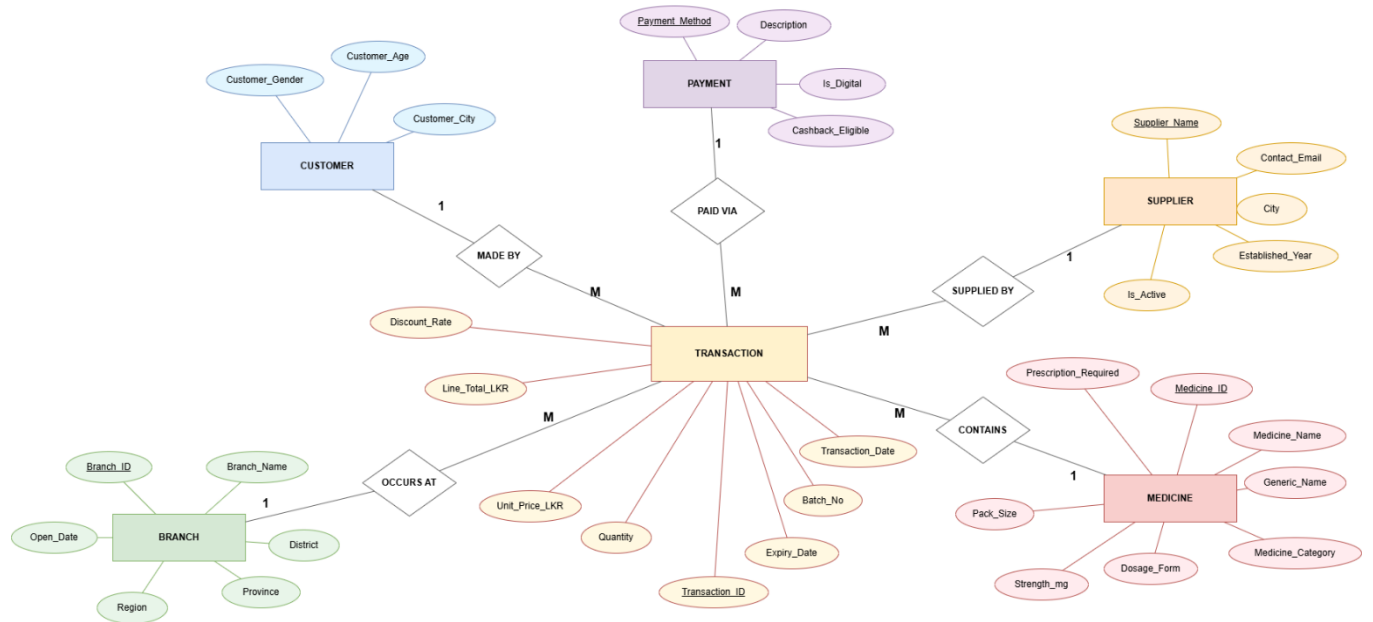
Entities in the Pharmacy OLTP Model

- **BRANCH** (Branch_ID PK, Branch_Name, District, Province) — Represents a pharmacy branch. Each branch can process multiple transactions.
- **MEDICINE** (Medicine_ID PK, Medicine_Name, Generic_Name, Medicine_Category, Dosage_Form, Strength_mg, Pack_Size) — Represents individual medicine products. A single medicine can be involved in multiple transactions.
- **SUPPLIER** (Supplier_Name PK, Prescription_Required) — Represents pharmaceutical suppliers. One supplier can supply multiple medicines.
- **CUSTOMER** (Customer_Gender, Customer_Age, Customer_City) — Represents anonymized customer segments based on demographics. Each segment can be associated with multiple transactions.
- **PAYMENT_METHOD** (Payment_Method PK) — Reference entity for payment types. One payment method can be used in multiple transactions.
- **TRANSACTION** (Transaction_ID PK, Transaction_Date, Branch_ID FK, Medicine_ID FK, Supplier_Name FK, Payment_Method FK, Quantity, Unit_Price_LKR, Discount_Rate, Line_Total_LKR) — Central transactional entity linking branches, medicines, suppliers, payment methods, and customers. Each transaction records the sale of a medicine and its associated details

Key Relationships

- **BRANCH – TRANSACTION (1:M)**: Each branch can handle many transactions. Every transaction is associated with exactly one branch.
- **MEDICINE – TRANSACTION (1:M)**: Each medicine can appear in multiple transactions, but each transaction records a specific medicine.
- **SUPPLIER – TRANSACTION (1:M)**: Each supplier can supply medicines involved in multiple transactions. Every transaction references specific supplier.
- **PAYMENT_METHOD – TRANSACTION (1:M)**: Each payment method can be used for multiple transactions. Every transaction is paid via a single payment method.
- **CUSTOMER – TRANSACTION (1:M)**: Each customer segment can be associated with many transactions. Every transaction is linked to exactly one customer segment.

Sri Lanka Pharmacy Sales — ER Diagram



[View ER Diagram](#)

Task 2: Preparation of Data Sources

2.1 Data Source Overview

To meet the assignment requirement of using at least two different data source types, the original single CSV file was split and enriched into four distinct source types. This arrangement simulates realistic multi-system environment where data arrives from different operational systems in a different formats

The table below shows each data source, its format, what kind of data it contains, and the staging table it is connected to.

Source	File / Object	Format	Content	Staging Table
Source 1	Source1_Transactions.csv	CSV (Flat File)	50,000 raw sales transaction records with financial measures and customer demographics	stg_Transactions
Source 2	BranchMaster table in PharmacySource DB	SQL Server Database (OLE DB)	7 pharmacy branch records including geographic data and opening dates	stg_Branches
Source 3 – Sheet 1	Medicine_Supplier_Reference.xlsx	Excel (.xlsx)	148 medicine reference records with categories, dosage forms, and strength details	stg_Medicines
Source 3 – Sheet 2	Medicine_Supplier_Reference.xlsx	Excel (.xlsx)	8 supplier records with contact details, city, established year, and active status	
Source 4	Source4_Payment_Methods.txt	Pipe-delimited Text File	3 payment method records with descriptions, digital flag, and cashback eligibility	

2.2 Source Type Details

2.2.1 CSV / Flat File Sources

File: Source1_Transactions.csv

The main transactional CSV file contains 50,000 rows extracted from the original OLTP dataset. It retains only the transactional columns, removing branch and medicine reference details that are managed in dedicated sources. This source feeds the central fact table pipeline

2.2.2 Excel Sources

File: Medicine_Supplier_Reference.xlsx (2 Sheets)

The medicine and supplier reference data is stored in a two-sheet Excel workbook. This simulates a business scenario where reference data is maintained in spreadsheets by the pharmaceutical procurement team..

2.2.3 SQL Server OLE DB Source

Database: PharmacySource | Table: dbo.BranchMaster

The branch master data is stored in a dedicated SQL Server database to simulate an operational source system. This source provides geographic reference data for all 7 pharmacy branches including district, province, region, and opening date — attributes that are not available in the transaction CSV.

2.2.4 Pipe-Delimited Text File (Payment Methods)

File: Source4_Payment_Methods.txt

The payment method reference data is stored as pipe-delimited text file. This source type demonstrates the ability to handle non-standard delimiters in SSIS, which is common real-world scenario when data is exported from legacy systems.

The file contains 3 records with columns sre Payment_Method, Description, Is_Digital, Cashback_Eligible, Processing_Days. In SSIS, this source uses a Flat File Connection Manager configured with the pipe character (|) as the column delimiter.

2.3 Data Enrichment and Hierarchy Planning

Several enrichment steps were applied to the source data to support analytical hierarchies:

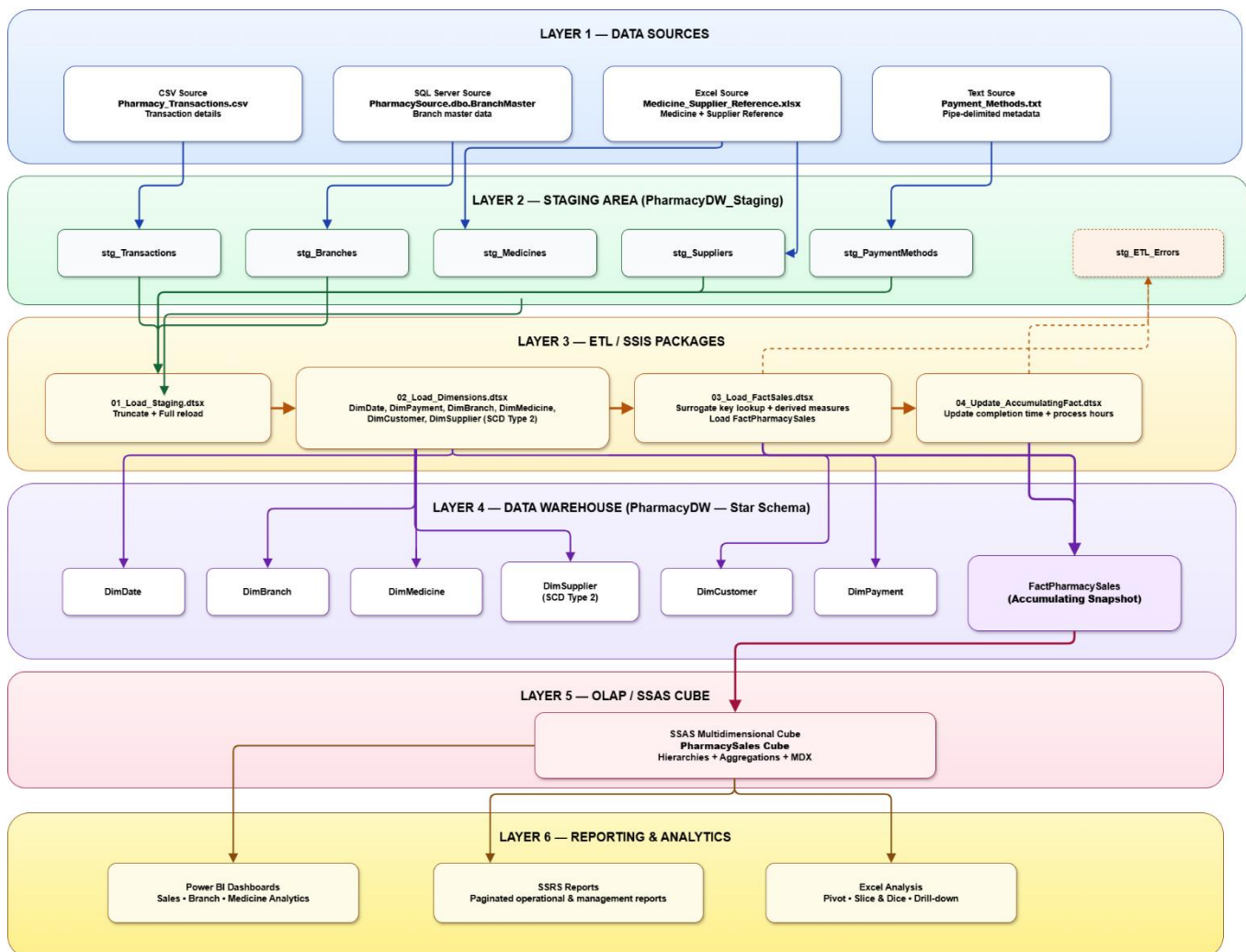
- **Geographic Hierarchy:** Province -> District -> BranchName — sourced from Branch Master (Source 2). This three-level hierarchy enables geographic drill-down analysis in SSAS.
- **Product Hierarchy:** MedicineCategory -> MedicineName — sourced from Medicine Reference (Source 3 Sheet 1). This two-level hierarchy supports product-level analytics.
- **Customer Segmentation:** Since no CustomerID exists, customers are grouped by AgeGroup (0-18, 19-35, 36-55, 56+) combined with Gender and City to form demographic segments. AgeGroup is a derived column computed in SSIS from the raw Customer_Age value.
- **Discount Measure:** DiscountAmount_LKR is a derived measure calculated as $\text{UnitPrice_LKR} \times \text{Quantity} \times \text{DiscountRate}$, added during the fact table ETL.

Task 3: Solution Architecture

3.1 Architecture Diagram

The DW & BI solution follows a classic Kimball-style layered pipeline architecture consisting of six functional layers. Data flows from heterogeneous operational sources through ETL processing into a dimensional data warehouse, and from there into analytical and reporting tools

Sri Lanka Pharmacy Data Warehouse Modern Solution Architecture



[View ER Diagram](#)

3.2 Component Descriptions

1. Data Sources Layer

This layer gathers raw data from four heterogeneous sources:

- Pharmacy_Transactions.csv (transactional data)
- PharmacySource.dbo.BranchMaster (branch master data)
- Medicine_Supplier_Reference.xlsx (medicine and supplier reference)
- Payment_Methods.txt (payment method metadata)
-

2. Staging Area (PharmacyDW_Staging)

A temporary landing zone in SQL Server that mirrors the structure of source systems. All extracted data is loaded here before transformation. The staging tables are truncated and fully reloaded in each ETL run to ensure data freshness and to isolate source systems from downstream processing.

3. ETL Layer (SSIS)

The transformation layer consists of four sequenced SSIS packages:

- 01_Load_Staging.dtsx: Truncates and loads all staging tables
- 02_Load_Dimensions.dtsx: Populates dimension tables (including SCD Type 2 for DimSupplier)
- 03_Load_FactSales.dtsx: Loads the fact table with surrogate key lookups and derived measures
- 04_Update_AccumulatingFact.dtsx: Updates accumulating snapshot columns

4. Data Warehouse

A SQL Server database implementing a classic star schema. It features one central accumulating snapshot fact table (FactPharmacySales) linked to six dimension tables (DimDate, DimBranch, DimMedicine, DimSupplier, DimCustomer, DimPayment) via surrogate keys. This design enables efficient analytical querying across multiple business dimensions.

5. OLAP Layer (SSAS)

SQL Server Analysis Services (SSAS) Multidimensional Cube (PharmacySales Cube) built on the data warehouse. It provides pre-aggregated data, defined hierarchies (e.g., Province -> District -> Branch, Medicine Category -> Medicine), and supports fast MDX-based operations including roll-up, drill-down, slice, dice, and pivot

6. Reporting & Analytics Layer

Power BI Dashboards: Interactive visualizations for sales, branch, and medicine analytics

SSRS Reports: Paginated operational and management reports

Task 4: Data Warehouse Design and Development

4.1 Dimensional Model Overview

The data warehouse uses a Star Schema design. This schema places one central fact table (FactPharmacySales) at the centre, connected to six surrounding dimension tables through foreign key relationships using surrogate keys. The star schema was chosen over snowflake because the dimension tables are relatively compact and denormalized, which simplifies ETL processing and SSAS cube design while optimising query performance.

4.2 Dimension and Fact Table Summary

Table	Type	Primary Key	Natural Key	Description
DimDate	Dimension	DateKey (INT)	FullDate	Calendar dimension generated for Jan 2022 – Jun 2023. Supports time-based hierarchies.
DimBranch	Dimension	BranchKey	BranchID	Geographic branch dimension with Province -> District -> Branch hierarchy.
DimMedicine	Dimension	MedicineKey	MedicineID	Product dimension with Medicine Category -> Medicine Name hierarchy.
DimSupplier	SCD Type 2 Dimension	SupplierKey	SupplierName	Supplier dimension tracking history for email and city changes over time.
DimCustomer	Dimension	CustomerKey	Gender+AgeGroup+City	Derived demographic segment dimension. No CustomerID in source — segmented by gender, age group, and city.
DimPayment	Dimension	PaymentKey	PaymentMethod	Payment method reference dimension with digital and cashback flags.
FactPharmacySales	Accumulating Fact	SalesKey	TransactionID	Central fact table at transaction grain. Holds 6 FK references

				and all financial measures.
--	--	--	--	-----------------------------

1.DimDate

DimDate is a standard calendar dimension populated by a SQL script rather than from source data. It stores one row per calendar date and contains attributes to support time-based analysis at any level of granularity. The DateKey column is an integer in the format YYYYMMDD (e.g., 20220115 for 15 January 2022).

```
SQLQuery1.... GF66 (69))* X
1
2  USE Pharmacy;
3  GO
4
5
6  CREATE TABLE DimDate (
7      DateKey      INT PRIMARY KEY,
8      FullDate     DATE, Year INT, Quarter INT, QuarterName NVARCHAR(10),
9      Month INT, MonthName NVARCHAR(20), Week INT, Day INT,
10     DayName NVARCHAR(15), DayOfWeek INT, IsWeekend BIT, IsHoliday BIT
11 );
12
```

2.DimBranch

DimBranch contains geographic information about each of the 7 pharmacy branches. The dimension includes a three-level geographic hierarchy: Province -> District -> BranchName. An additional Region attribute was introduced during enrichment to support a higher-level grouping above Province. The OpenDate attribute enables analysis of branch performance relative to its age.

```
SQLQuery1.... GF66 (69))* X
1
2  USE Pharmacy;
3  GO
4  CREATE TABLE DimBranch (
5      BranchKey    INT IDENTITY(1,1) PRIMARY KEY,
6      BranchID     INT,
7      BranchName   NVARCHAR(100),
8      District     NVARCHAR(50),
9      Province     NVARCHAR(50),
10     Region       NVARCHAR(50),
11     OpenDate     DATE
12 );
```

3. DimMedicine

DimMedicine stores reference information for all 148 unique medicines. The Medicine_Category attribute introduces a two-level product hierarchy: MedicineCategory → MedicineName. The DosageForm attribute (Tablet, Capsule, Syrup, Cream, Inhaler, Drops, Ointment, Sachet) provides an additional analytical dimension.

```
SQLQuery1..... GF66 (69))* ×
1
2  USE Pharmacy;
3  GO
4  -- DimMedicine
5  CREATE TABLE DimMedicine (
6      MedicineKey      INT IDENTITY(1,1) PRIMARY KEY,
7      MedicineID       INT,
8      MedicineName     NVARCHAR(150),
9      GenericName      NVARCHAR(100),
10     MedicineCategory NVARCHAR(50),
11     DosageForm        NVARCHAR(50),
12     StrengthMg        DECIMAL(8,2),
13     PackSize         INT
14 );
```

4. DimSupplier — SCD Type 2

DimSupplier is the Slowly Changing Dimension (Type 2) in this solution. It stores details for all 8 pharmaceutical suppliers. The ContactEmail and City columns are designated as historical (Type 2) attributes because they may change when a supplier updates their contact information or relocates. When a change is detected, the old row is expired (IsCurrent = 0, ExpiryDate = today) and a new row is inserted with the updated values (IsCurrent = 1, EffectiveDate = today, ExpiryDate = 9999-12-31).

```
SQLQuery1..... GF66 (69))* ×
1
2  USE Pharmacy;
3  GO
4
5  CREATE TABLE DimSupplier (
6      SupplierKey      INT IDENTITY(1,1) PRIMARY KEY,
7      SupplierName     NVARCHAR(100),
8      ContactEmail     NVARCHAR(100),
9      City             NVARCHAR(50),
10     EstablishedYear   INT,
11     IsActive         BIT,
12     EffectiveDate     DATE,
13     ExpiryDate        DATE,
14     IsCurrent        BIT DEFAULT 1
15 );
```

5. DimCustomer

Since the source data contains no CustomerID, customers cannot be tracked individually. Instead, a demographic segment dimension is created by deriving the AgeGroup from Customer_Age using SSIS Derived Column transformation, then grouping distinct combinations of Gender, AgeGroup, and City. This creates 64 unique customer segments which are loaded into DimCustomer.

Age Group	Age Range
0-18	0 to 18 years
19-35	19 to 35 years
36-55	36 to 55 years
56+	56 years and above

```
SQLQuery1.... GF66 (69))* ✕
1
2  USE Pharmacy;
3  GO
4
5  CREATE TABLE DimCustomer (
6      CustomerKey INT IDENTITY(1,1) PRIMARY KEY,
7      Gender NVARCHAR(10),
8      AgeGroup NVARCHAR(20),
9      City NVARCHAR(50)
10 );
11
```

6. DimPayment

DimPayment stores the three payment method types available in the pharmacy system (Card, Cash, Insurance). It includes enriched attributes: Description (full description of the payment type), IsDigital (whether the payment is digital or physical), and CashbackEligible (whether cashback applies). These attributes were added from the pipe-delimited text file source.

```
SQLQuery1.... GF66 (69))* ✕
1
2  USE Pharmacy;
3  GO
4
5  CREATE TABLE DimPayment (
6      PaymentKey INT IDENTITY(1,1) PRIMARY KEY,
7      PaymentMethod NVARCHAR(20),
8      Description NVARCHAR(200),
9      IsDigital BIT,
10     CashbackEligible BIT
11 );
```


7. FactPharmacySales

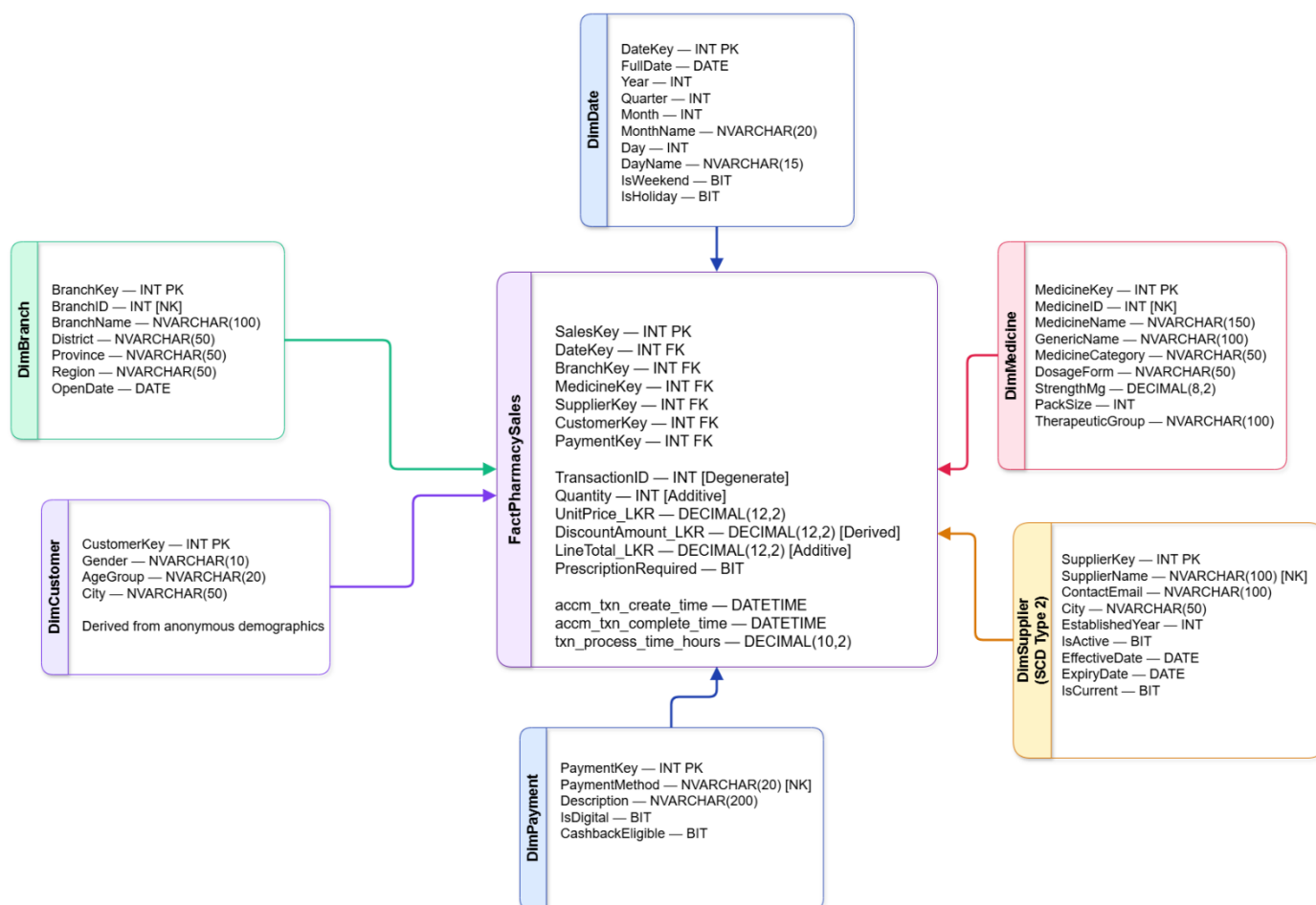
FactPharmacySales is the central accumulating fact table. Each row represents one unique pharmacy sales transaction. The table holds surrogate foreign keys to all six dimension tables and includes the following measures.

```
SQLQuery1.... GF66 (69))* ✕ ×
1
2  USE Pharmacy;
3  GO
4
5  -- FactPharmacySales
6  CREATE TABLE FactPharmacySales (
7      SalesKey          INT IDENTITY(1,1) PRIMARY KEY,
8      DateKey           INT REFERENCES DimDate(DateKey),
9      BranchKey         INT REFERENCES DimBranch(BranchKey),
10     MedicineKey       INT REFERENCES DimMedicine(MedicineKey),
11     SupplierKey       INT REFERENCES DimSupplier(SupplierKey),
12     CustomerKey       INT REFERENCES DimCustomer(CustomerKey),
13     PaymentKey        INT REFERENCES DimPayment(PaymentKey),
14     TransactionID     INT,
15     Quantity          INT,
16     UnitPrice_LKR     DECIMAL(12,2),
17     DiscountRate      DECIMAL(5,4),
18     DiscountAmount_LKR DECIMAL(12,2),
19     LineTotal_LKR     DECIMAL(12,2),
20     PrescriptionRequired BIT,
21     accm_txn_create_time DATETIME,
22     accm_txn_complete_time DATETIME,
23     txn_process_time_hours DECIMAL(10,2)
24
```

4.3 Data Warehouse Schema Diagram (Star Schema)

The PharmacyDW Star Schema is a dimensional data model designed for efficient analytical querying and reporting. It consists of a central accumulating snapshot fact table (FactPharmacySales) connected to six dimension tables — DimDate, DimBranch, DimMedicine, DimSupplier (SCD Type 2), DimCustomer, and DimPayment — through surrogate keys.

PharmacyDW — Star Schema Dimensional Model



[\(View star diagram\)](#)

4.4 Design Assumptions

- Customer anonymity: No CustomerID exists in the source data. Customers are segmented by Gender + AgeGroup + City combinations. This is documented as a design assumption in the fact table where CustomerKey references a demographic segment rather than an individual.
- DimSupplier SCD Type 2: Supplier ContactEmail and City are treated as historically tracked attributes. If either changes, the old row is expired and a new row is inserted to preserve history for accurate supplier performance analysis.
- DimDate range: The date dimension is generated for the exact data range of Jan 2022 to Jun 2023. This covers all 546 distinct transaction dates.
- DiscountAmount_LKR is a derived measure calculated in SSIS as: $\text{UnitPrice_LKR} \times \text{Quantity} \times \text{DiscountRate}$.
- The TransactionID is stored as a degenerate dimension in the fact table — it has no associated dimension table but is retained for traceability.
- accm_txn_create_time is set to GETDATE() during the initial fact load, not the original transaction date, to demonstrate the accumulating fact pattern as required by Task 6.

Task 5: ETL Development

5.1 ETL Design Overview

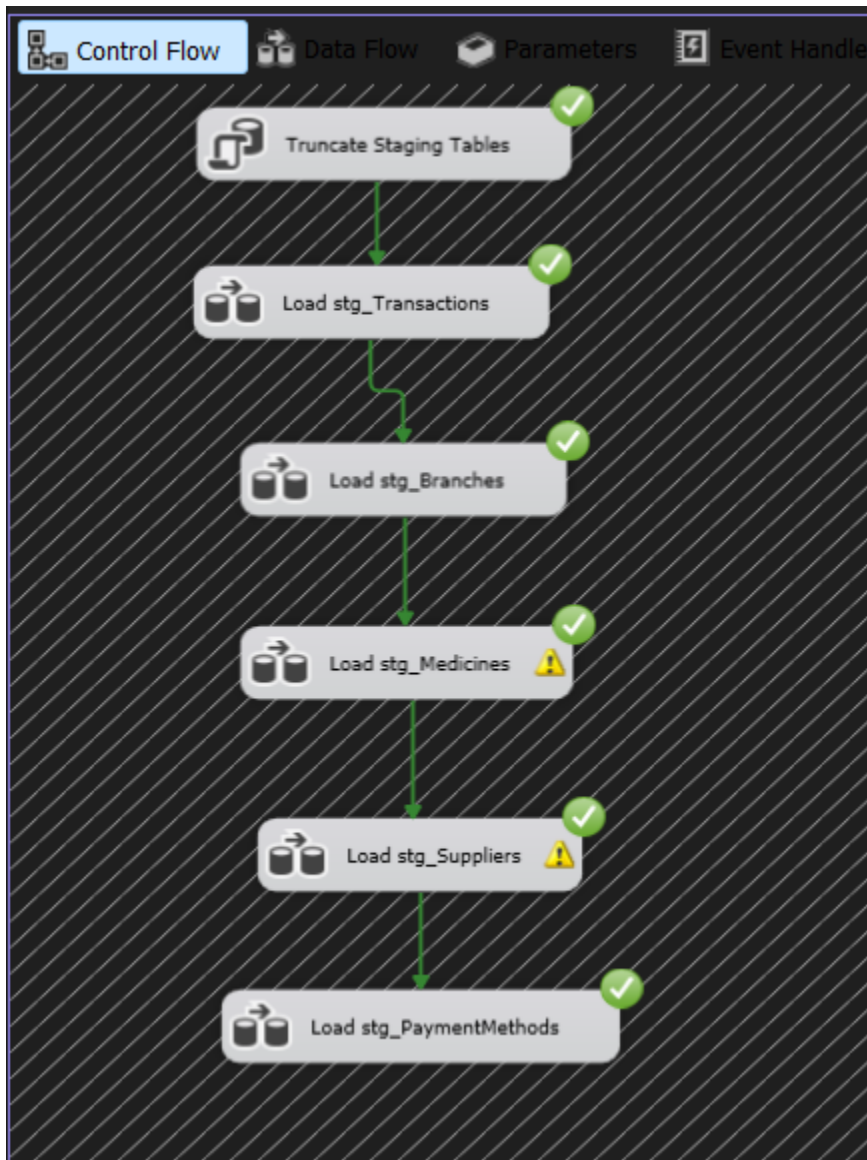
The entire ETL process was built using SQL Server Integration Services (SSIS) in Visual Studio. Three sequential SSIS packages were created to load data from the four source types into the data warehouse. To maintain referential integrity, dimension tables must always be loaded before the fact table, as the fact table's lookup transformations depend on surrogate keys already existing in the dimension tables.

Package	Name	Purpose	Execution Order
Package 1	01_Load_Staging.dtsx	Extracts all 4 source types into 5 staging tables	1st — always run first
Package 2	02_Load_Dimensions.dtsx	Loads all 6 dimension tables from staging including SCD Type 2 for DimSupplier	2nd — after staging
Package 3	03_Load_FactSales.dtsx	Loads FactPharmacySales using lookup transformations to resolve surrogate keys	3rd — after dimensions
Package 4	04_Update_AccumulatingFact.dtsx	Updates accm_txn_complete_time and	4th — runs separately

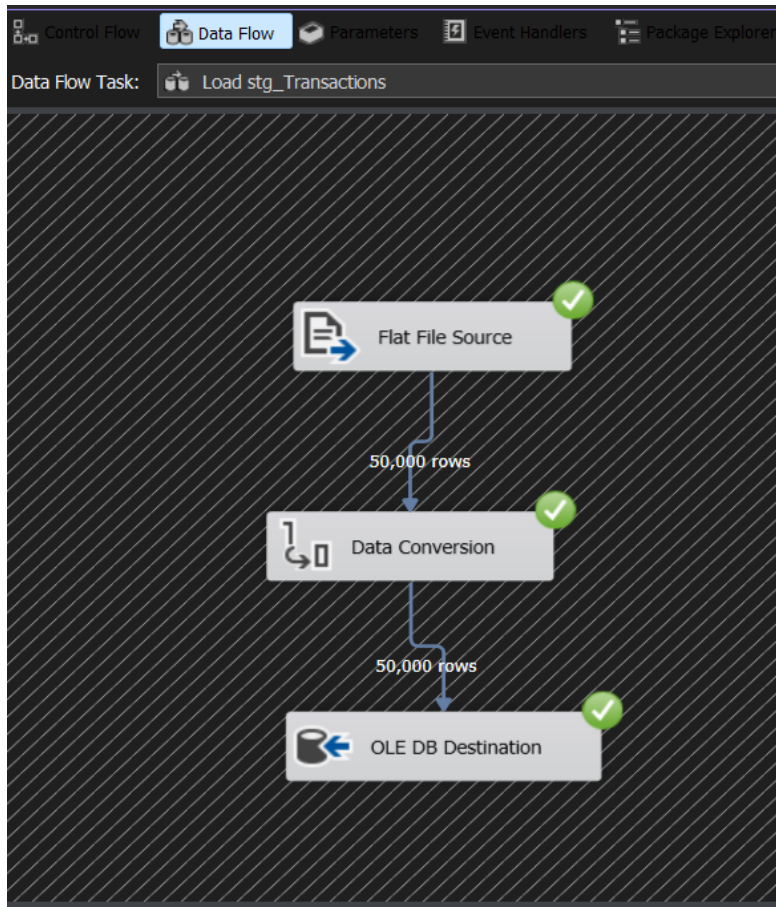
	txn_process_time_hours in the fact table	
--	---	--

5.2 Staging Layer ETL

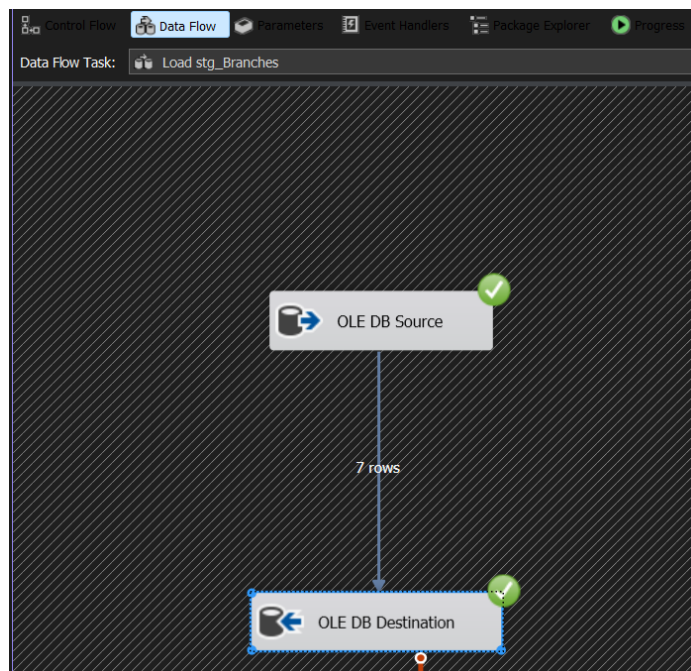
Package 1 loads all raw data from the four source types into the five staging tables. The Control Flow begins with an Execute SQL Task that truncates all staging tables, followed by five Data Flow Tasks chained with precedence constraints (green arrows) to enforce sequential execution.



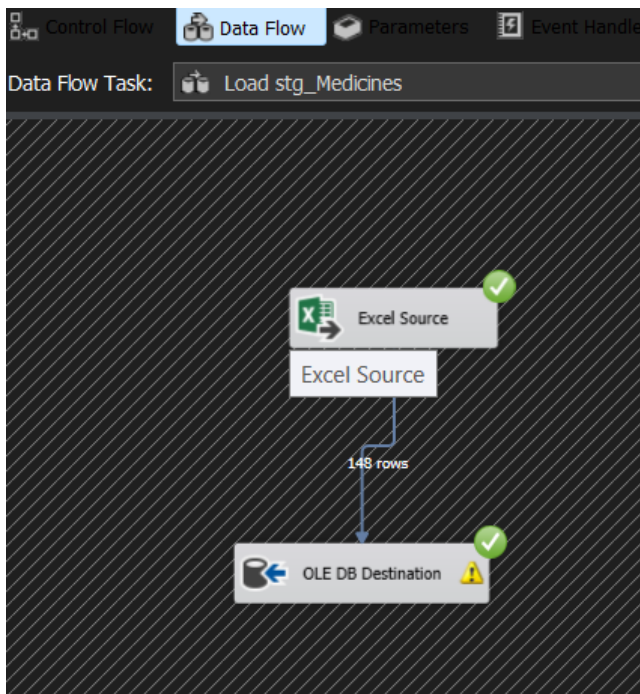
1. Load stg_Transactions (CSV Source)



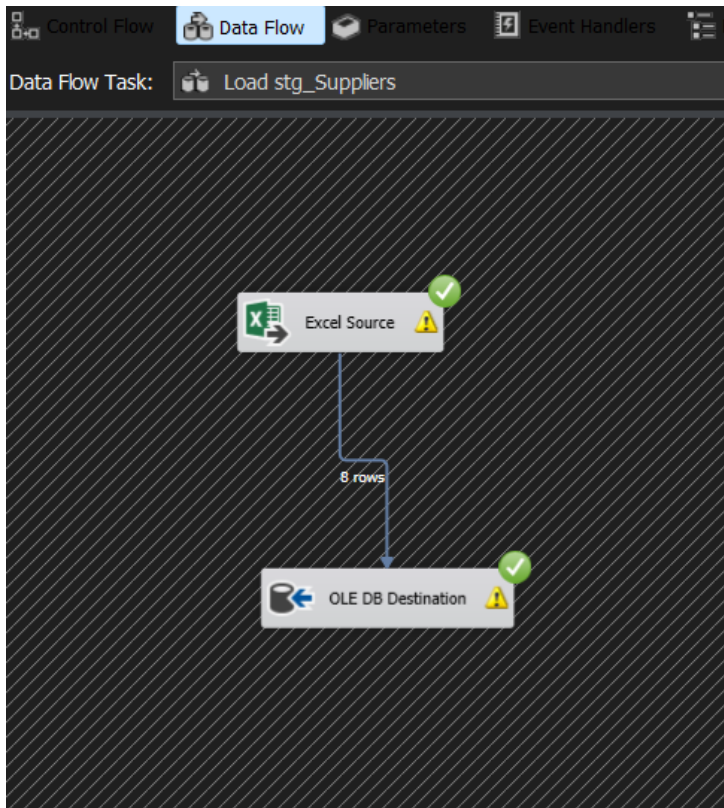
2. Load stg_Branches (SQL Server DB Source)



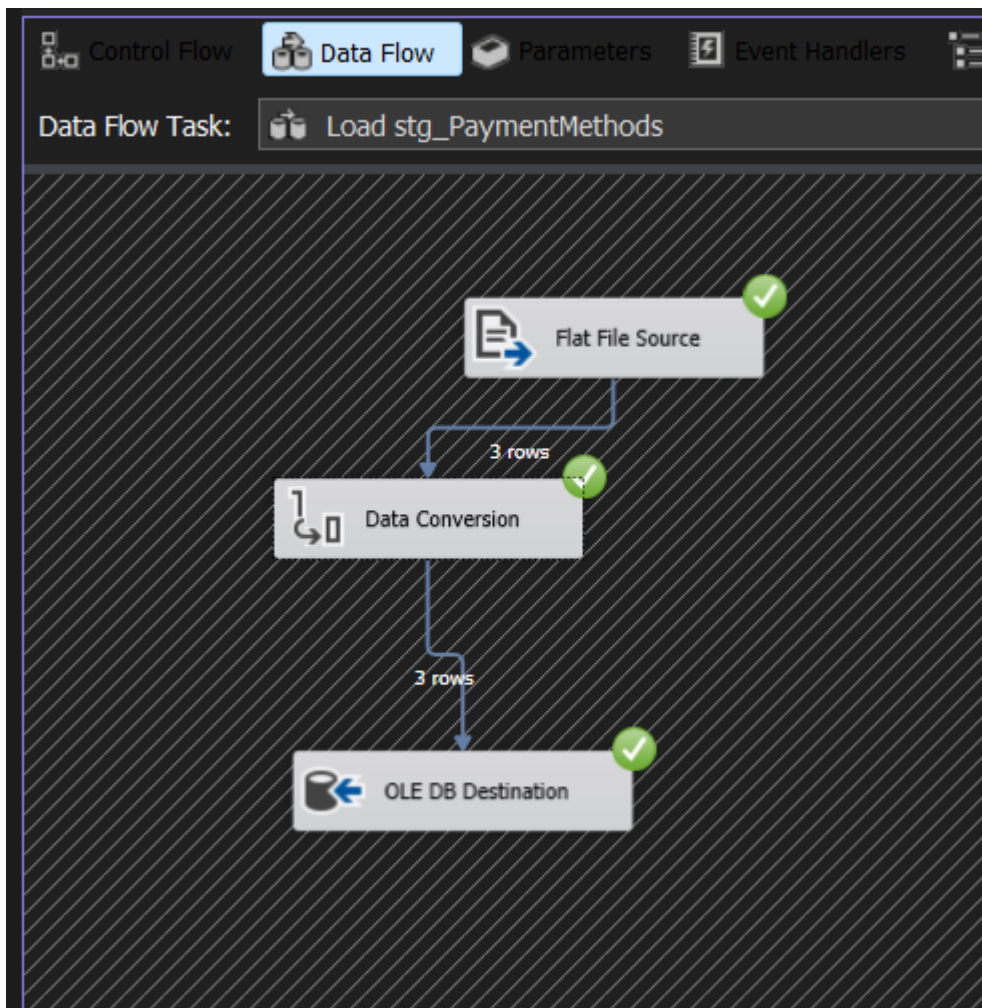
3. Load stg_Medicines (Excel Source, Sheet 1).



4. Load stg_Suppliers (Excel Source, Sheet 2)



5. Load stg_PaymentMethods (Pipe-Delimited Text)



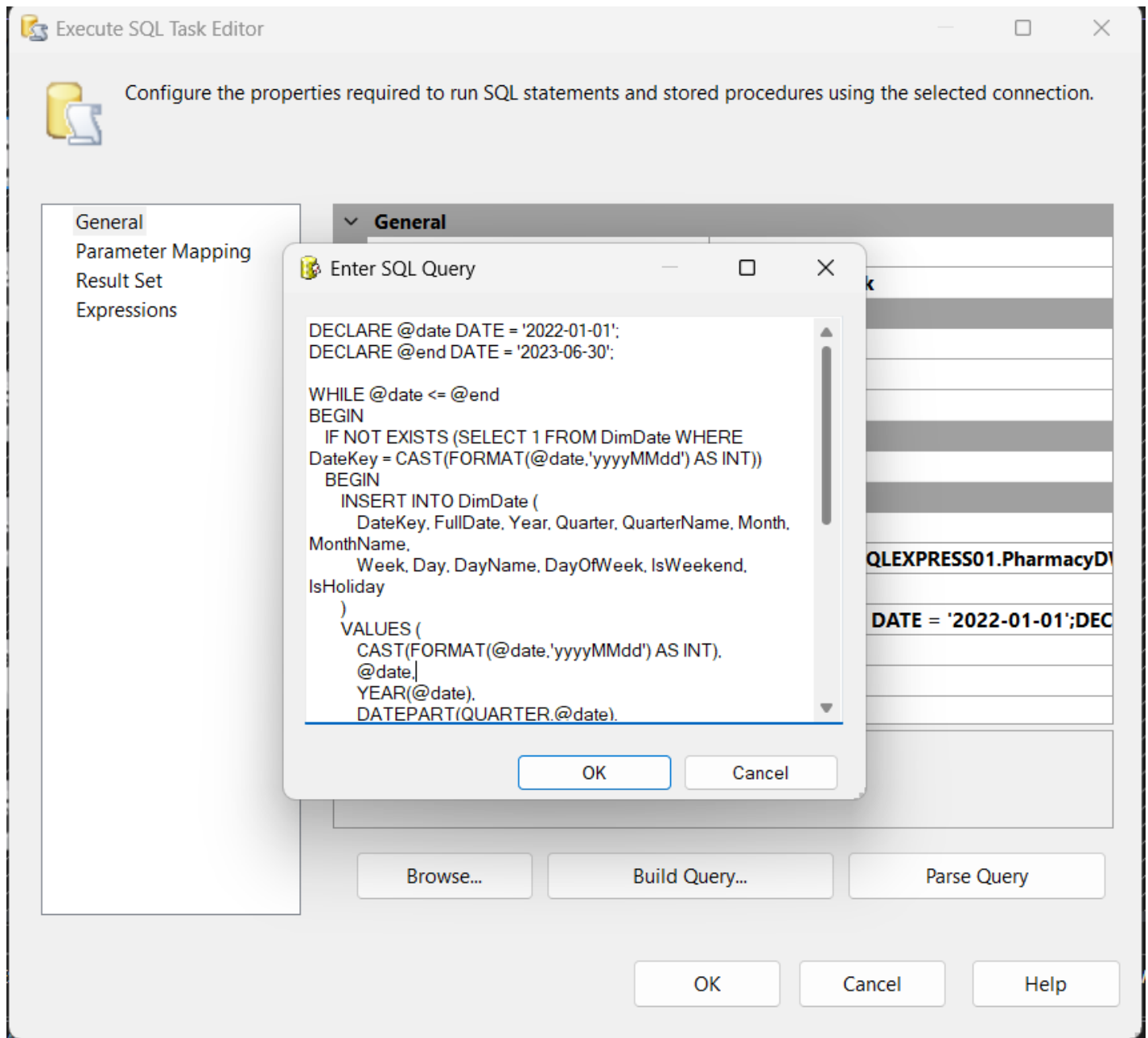
5.3 Dimension ETL

The second SSIS package, 02_Load_Dimensions.dtsx, is responsible for populating all six dimension tables in the PharmacyDW star schema. It executes in a strict sequential order: first loading DimDate using an Execute SQL Task, followed by DimPayment, DimBranch, DimMedicine, DimCustomer (with AgeGroup derivation), and finally DimSupplier using Slowly Changing Dimension (SCD Type 2) transformation to track changes in ContactEmail and City. This package ensures that all dimension surrogate keys are generated and available before the fact table load, maintaining referential integrity. Proper precedence constraints are used to enforce the loading sequence, and error handling is implemented to log any data quality issues during dimension population.

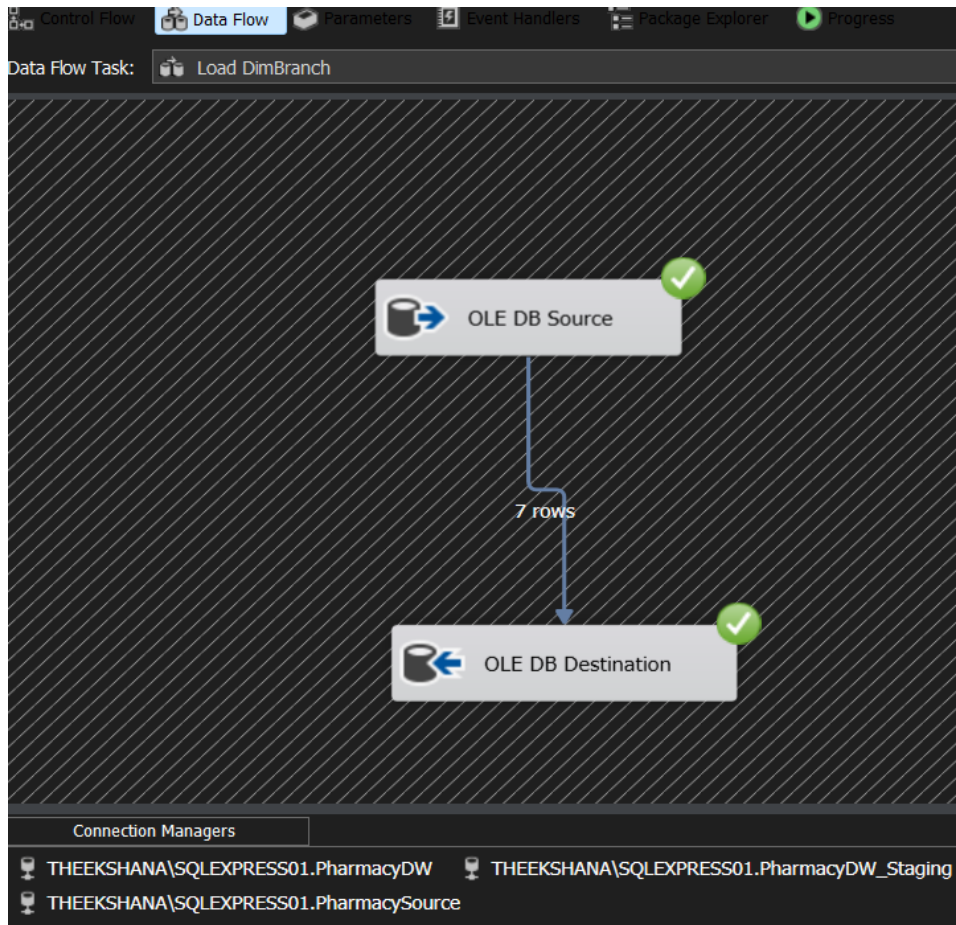


1. DimDate Load

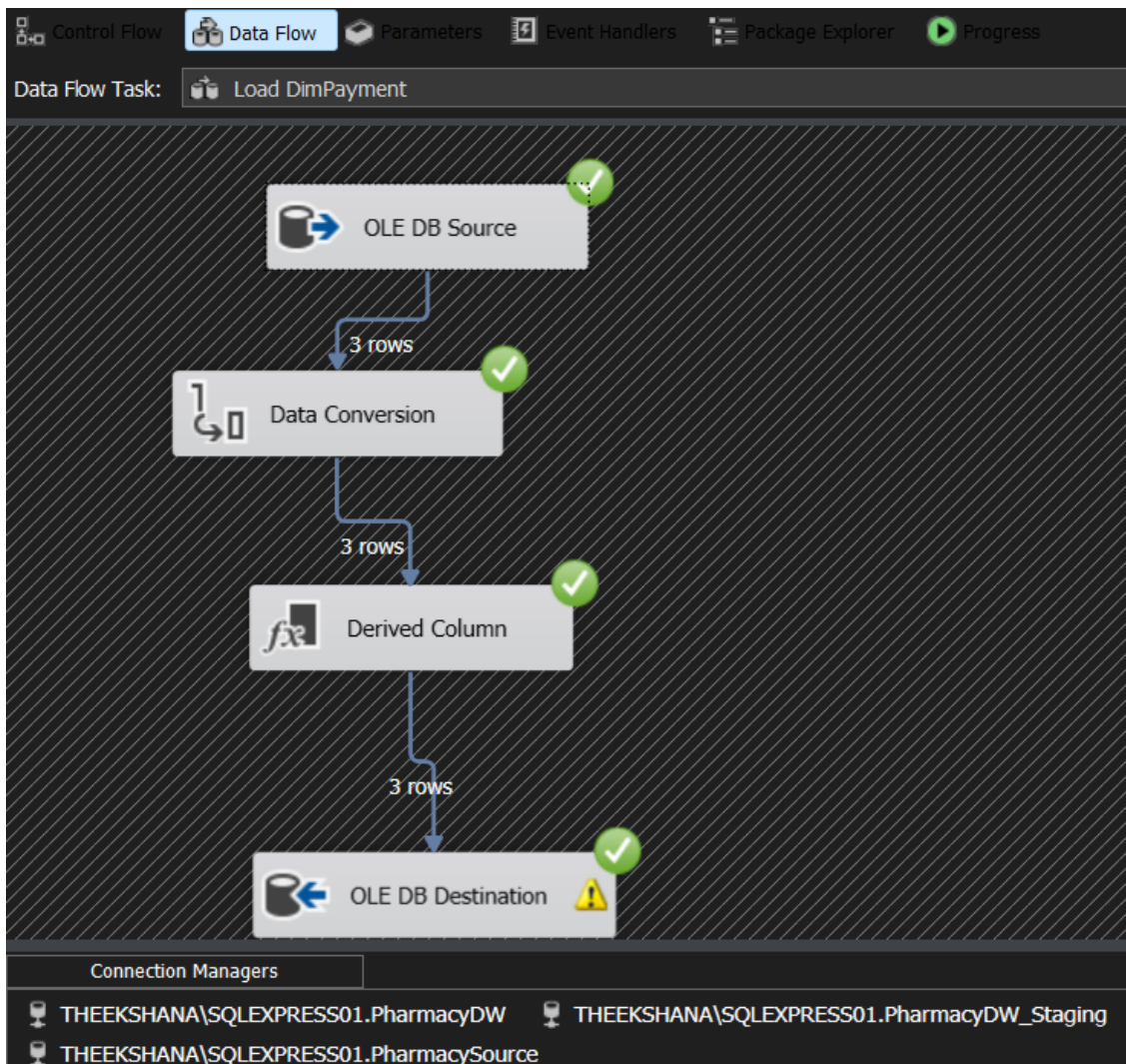
DimDate is a standard time dimension table that provides temporal context for all sales analysis. It contains attributes such as FullDate, Year, Quarter, MonthName, DayName, IsWeekend, and IsHoliday, enabling time-based roll-up, drill-down, and trend analysis. In the SSIS package, this dimension is populated first using an Execute SQL Task with a date generation script covering the period from January 2022 to June 2023.



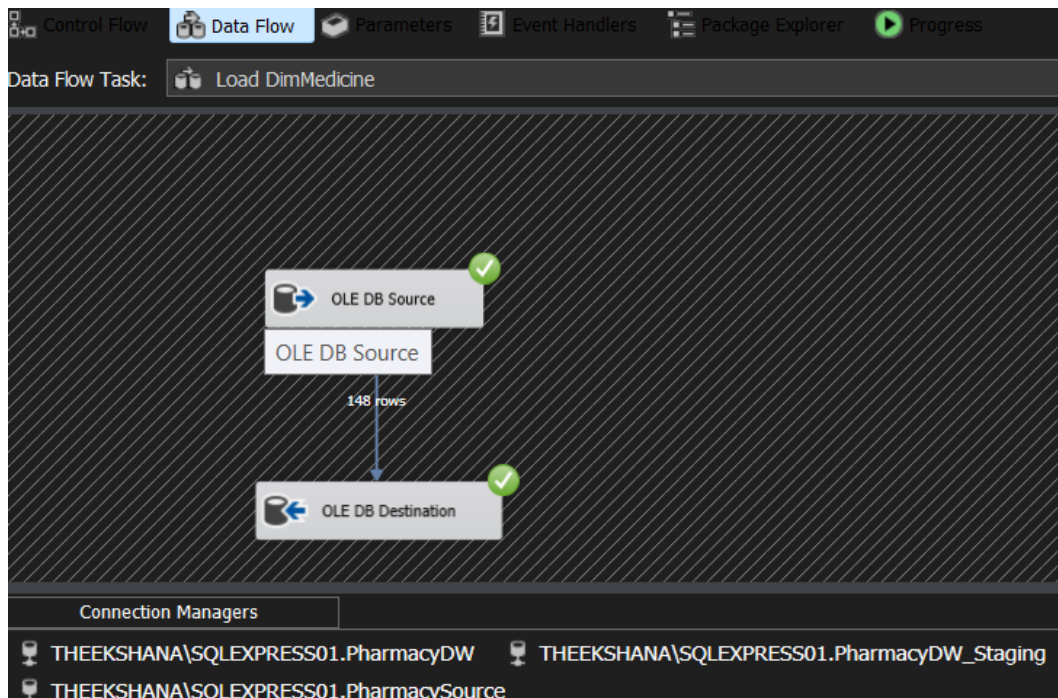
2. DimBranch Load



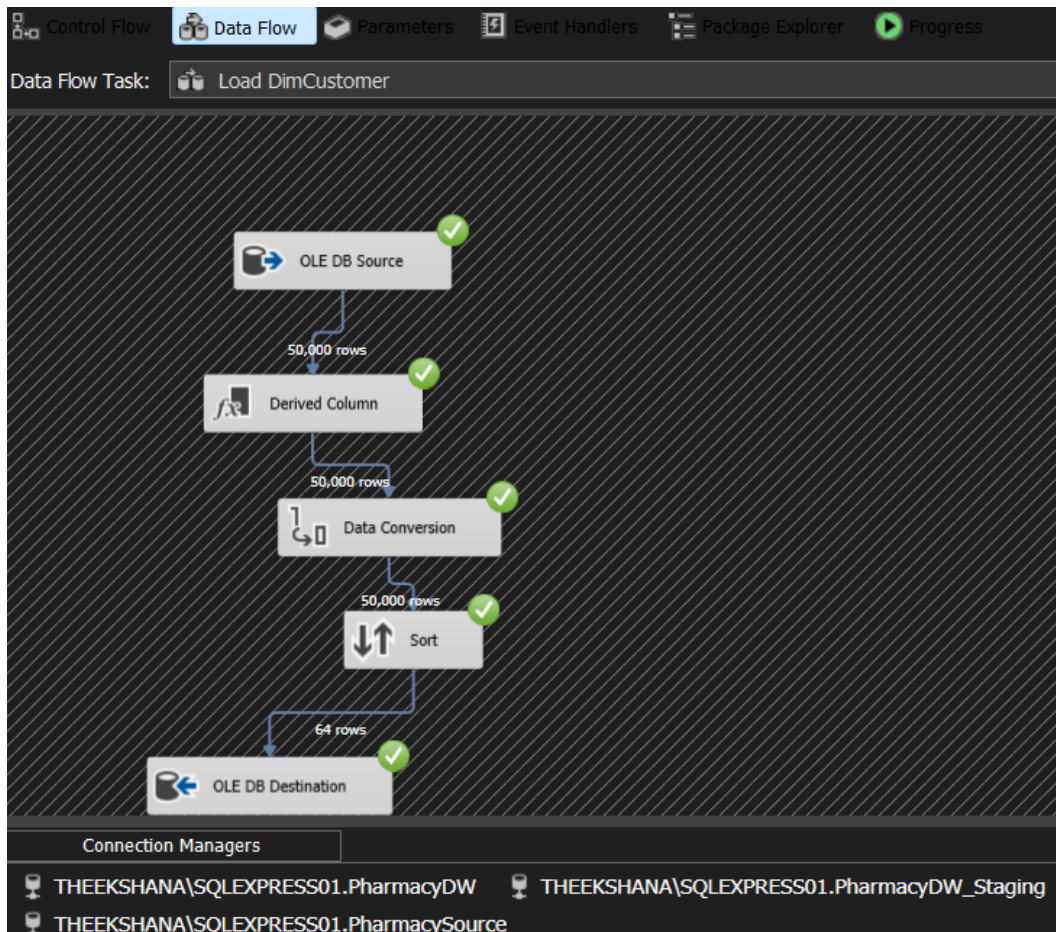
3. DimPayment Load



4. DimMedicine Load



5. DimCustomer Load



6. DimSupplier (SCD Type 2)

DimSupplier is implemented as a Slowly Changing Dimension (Type 2) to track historical changes in supplier details such as ContactEmail and City. In the SSIS package 02_Load_Dimensions.dtsx, the Slowly Changing Dimension transformation is used with SupplierName as the business key, automatically creating new versions of supplier records when changes occur while maintaining history through EffectiveDate, ExpiryDate, and IsCurrent columns.

The screenshot shows the 'Select a Dimension Table and Keys' step of the Slowly Changing Dimension Wizard. The connection manager is set to 'THEEKSHANA\SQLEXPRESS01.PharmacyDW'. The table or view selected is '[dbo].[DimSupplier]'. Below this, a table lists the input columns and their corresponding dimension columns and key types.

Input Columns	Dimension Columns	Key Type
City	City	Not a key column
ContactEmail	ContactEmail	Not a key column
	EffectiveDate	
EstablishedY...	EstablishedYear	Not a key column
	ExpiryDate	
IsActive	IsActive	Not a key column

Navigation buttons at the bottom include: Help, < Back, Next >, Finish >>|, and Cancel.

The screenshot shows the 'Slowly Changing Dimension Columns' step of the Slowly Changing Dimension Wizard. It provides instructions on how to manage changes to column data by setting the change type for each dimension column. The 'City' column is currently selected.

Fixed Attribute
Select this type when the value in a column should not change. Changes are treated as errors.

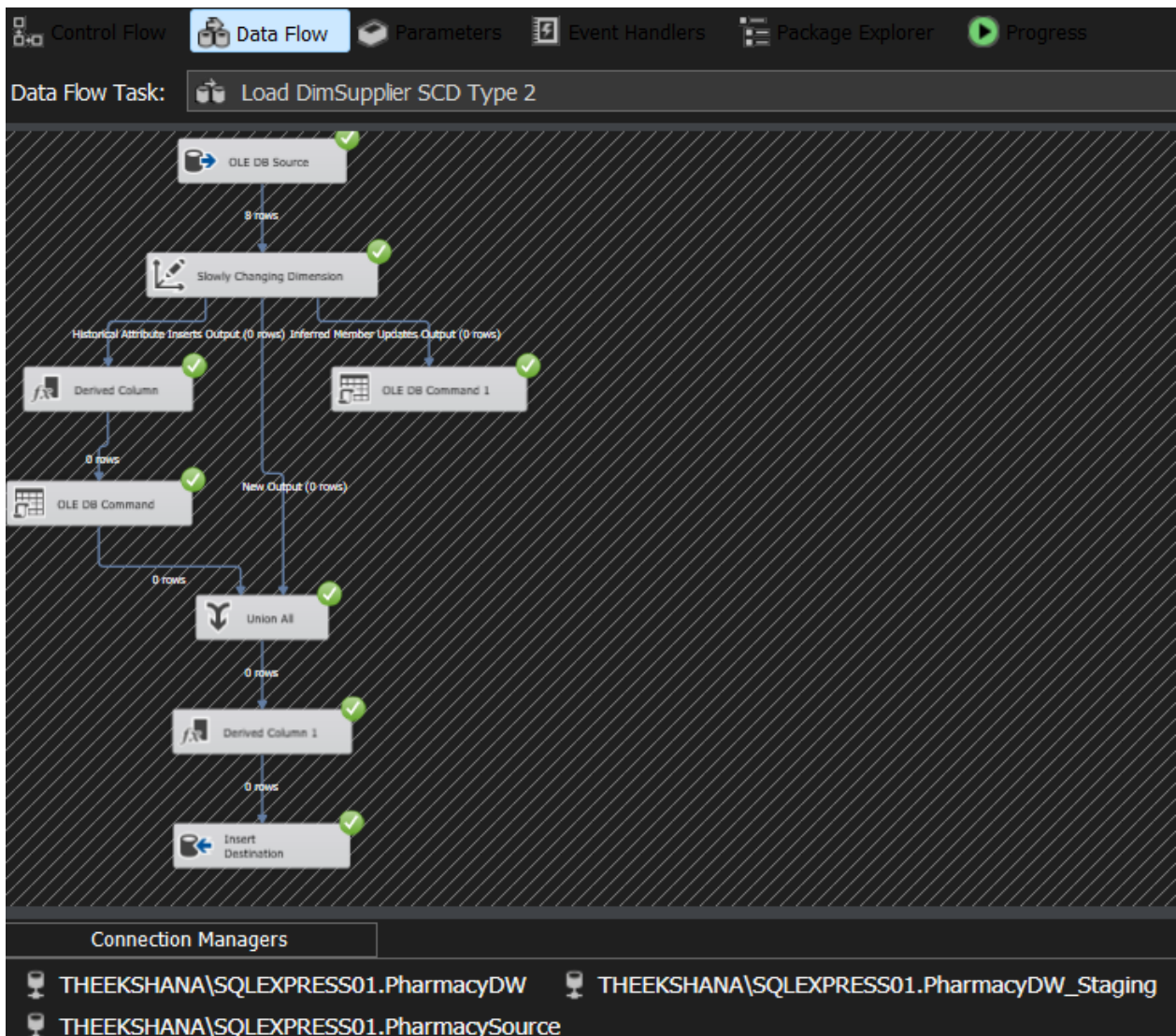
Changing Attribute
Select this type when changed values should overwrite existing values. This is a Type 1 change.

Historical Attribute
Select this type when changes in column values are saved in new

Select a change type for slowly changing dimension columns:

Dimension Columns	Change Type
City	Historical a...
ContactEmail	Historical a...
EstablishedYear	Fixed attrib...
IsActive	Fixed attrib...

Navigation buttons at the bottom include: Help, < Back, Next >, Finish >>|, and Cancel.



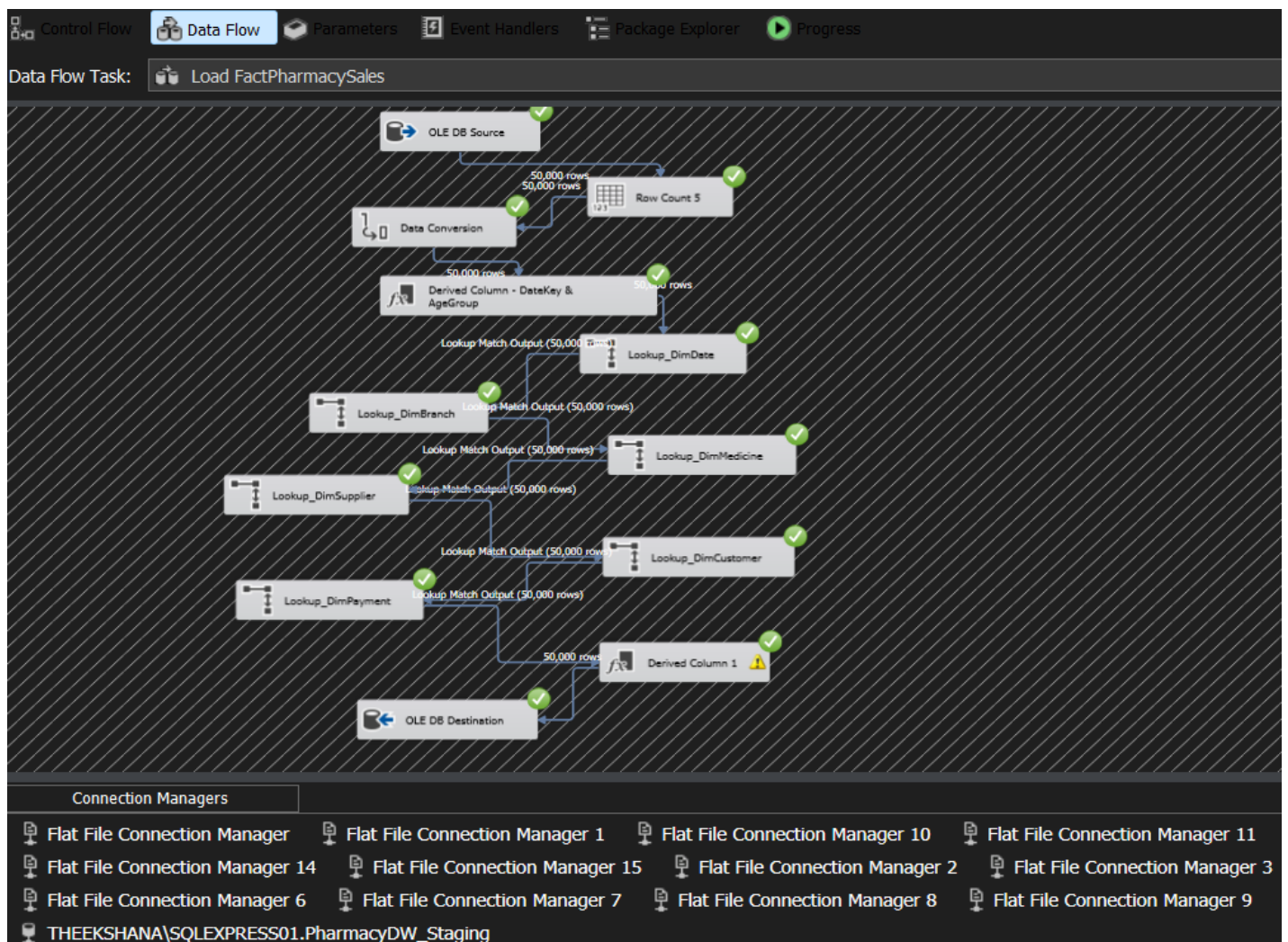
5.4 Fact Table ETL (FactPharmacySales)

It reads 50,000 transaction rows from staging and resolves surrogate keys from all six dimensions using Lookup transformations, then loads the resolved rows into FactPharmacySales.

The Data Flow pipeline in detail:

1. OLE DB Source — reads 50,000 rows from stg_Transactions
2. Data Conversion — converts Transaction_Date string to DT_DBDATE
3. Derived Column — computes DateKey = (YEAR×10000)+(MONTH×100)+DAY and AgeGroup from Customer_Age
4. Lookup_DimDate — matches DateKey → returns DateKey_Out (INT)
5. Lookup_DimBranch — matches Branch_ID → BranchID → returns BranchKey
6. Lookup_DimMedicine — matches Medicine_ID → MedicineID → returns MedicineKey

7. Lookup_DimSupplier — matches Supplier_Name → SupplierName (WHERE IsCurrent=1) → returns SupplierKey
8. Lookup_DimCustomer — matches Gender + AgeGroup + City → returns CustomerKey
9. Lookup_DimPayment — matches Payment_Method → PaymentMethod → returns PaymentKey
10. Derived Column — computes DiscountAmount_LKR = UnitPrice_LKR × Quantity × DiscountRate and accm_txn_create_time = GETDATE()
11. OLE DB Destination — maps all columns to FactPharmacySales



5.5 Validation

SQLQuery3.sq... GF66 (106) SQLQuery2.sq... GF66 (101) SQLQuery1.... GF66 (68))* ✕

```

1  USE PharmacyDW;
2  SELECT COUNT(*) AS TotalFactRows FROM FactPharmacySales; -- Expected: 50000
3
4  SELECT TOP 5 f.TransactionID, d.FullDate, b.BranchName,
5             m.MedicineName, s.SupplierName, p.PaymentMethod,
6             f.Quantity, f.LineTotal_LKR, f.accm_txn_create_time
7  FROM FactPharmacySales f
8  JOIN DimDate      d ON f.DateKey   = d.DateKey
9  JOIN DimBranch    b ON f.BranchKey = b.BranchKey
10 JOIN DimMedicine  m ON f.MedicineKey = m.MedicineKey
11 JOIN DimSupplier  s ON f.SupplierKey = s.SupplierKey
12 JOIN DimPayment   p ON f.PaymentKey = p.PaymentKey;

```

119 % No issues found Ln

Results Messages

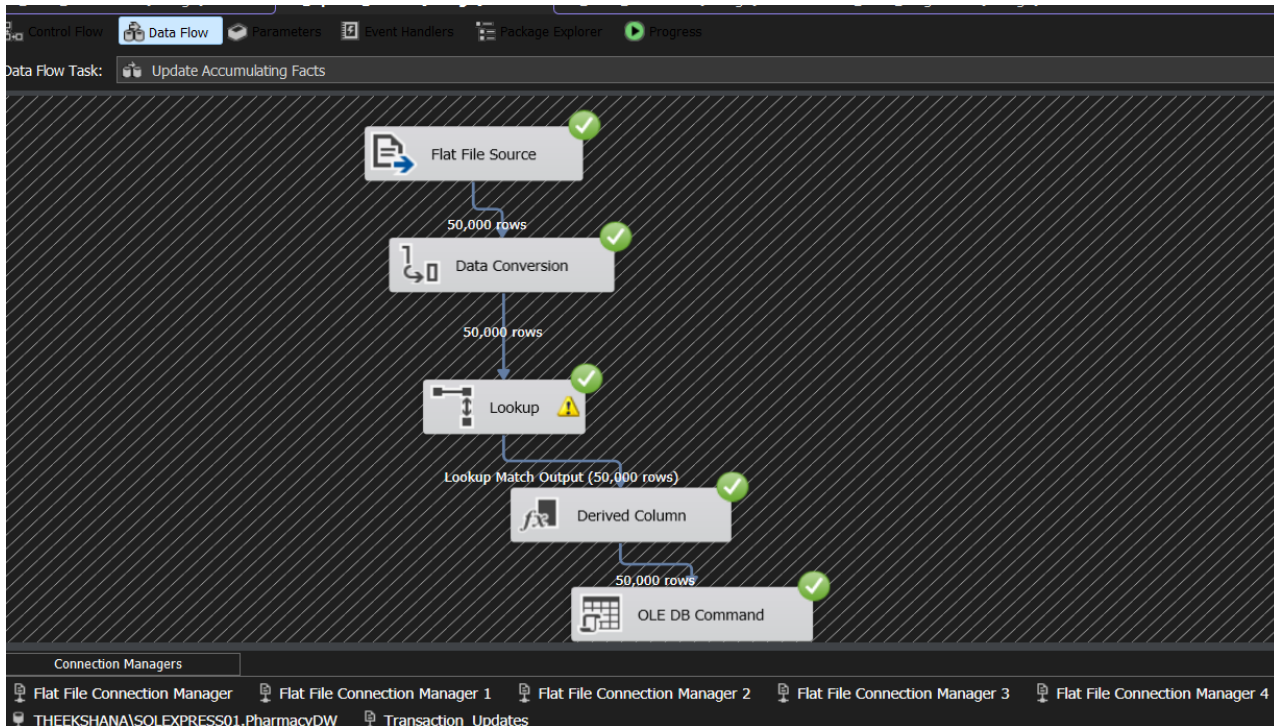
	TotalFactRows
1	50000

	TransactionID	FullDate	BranchName	MedicineName	SupplierName	PaymentMethod	Quantity	LineTotal_LKR	accm_txn_create_time
1	126	2023-04-17	Pharmacy - Ratnapura	Panadol Acyclovir Sachet	Ceylon Pharma Traders	Card	2	3845.33	2026-03-30 18:26:02.767
2	110	2022-06-06	Pharmacy - Galle	Cardio Chlorhexidine 500mg Drops	Ceylon Pharma Traders	Card	1	3282.40	2026-03-30 18:26:02.767
3	31	2023-04-10	Pharmacy - Negombo	Vita Glucalazide Sachet	Softlogic Pharma Distribution	Insurance	5	2583.90	2026-03-30 18:26:02.767
4	112	2022-10-30	Pharmacy - Negombo	Panadol Amoxicillin + Clavulanate Sachet	Astron Pharma Supply	Cash	5	12806.29	2026-03-30 18:26:02.767
5	117	2022-02-12	Pharmacy - Kurunegala	Cardio Doxycycline Syrup 100ml	Softlogic Pharma Distribution	Card	3	1580.24	2026-03-30 18:26:02.767

Task 6: ETL Development – Accumulating Fact Tables

6.1 Overview of Accumulating Fact Tables

An accumulating snapshot fact table tracks the lifecycle of a business process that has a defined start and end. In this pharmacy sales data warehouse, FactPharmacySales is designed as an accumulating fact table where each sales transaction has a creation time (when the record was first loaded) and a completion time (when the transaction was confirmed as fully processed by the supplier and branch systems).



6.2 Fact Table Extension

The FactPharmacySales table was extended with three columns to support the accumulating snapshot fact pattern:

- **accm_txn_create_time (DATETIME)**

This column is populated during the initial ETL process using GETDATE(). It represents the starting point of the accumulating snapshot by recording when the transaction enters the data warehouse.

- **accm_txn_complete_time (DATETIME, nullable)**

This column is updated later through a separate SSIS package after matching the TransactionID. It represents the completion point of the transaction, enabling tracking of the full transaction lifecycle.

- **txn_process_time_hours (DECIMAL(10,2), nullable)**

This value is calculated using the DATEDIFF function between the creation and completion timestamps. It is used to measure the processing time of a transaction and supports performance analysis.

6.3 Completion Dataset

A CSV file named Transaction_Completion_Updates.csv was prepared to simulate transaction completion events arriving from an external system. The file structure matches the requirement specified in the assignment brief

```
TransactionID,accm_txn_complete_time
41,2026-03-29 06:44:47
141,2026-03-27 14:44:47
49,2026-03-31 16:44:47
80,2026-03-30 09:44:47
112,2026-03-28 00:44:47
72,2026-03-27 22:44:47
99,2026-03-31 04:44:47
125,2026-03-26 23:44:47
79,2026-03-27 14:44:47
135,2026-03-31 10:44:47
103,2026-03-27 02:44:47
7,2026-03-30 16:44:47
37,2026-03-29 22:44:47
113,2026-03-30 10:44:47
136,2026-03-31 09:44:47
139,2026-03-27 08:44:47
64,2026-03-28 17:44:47
2,2026-03-30 23:44:47
10,2026-03-29 18:44:47
74,2026-03-30 01:44:47
138,2026-03-30 04:44:47
17,2026-03-30 21:44:47
62,2026-03-28 22:44:47
46,2026-03-29 04:44:47
129,2026-03-31 14:44:47
47,2026-03-27 08:44:47
```

6.4 Error Tables

An error table was created in PharmacyDW_Staging to capture rows that did not match any existing transaction in FactPharmacySales during the update process.

```

1 CREATE TABLE stg_ETL_Errors
2 (
3     [ErrorID] INT IDENTITY(1,1) NOT NULL,
4     [ErrorTime] DATETIME NOT NULL,
5     [ErrorSource] NVARCHAR(255) NOT NULL,
6     [ErrorMessage] NVARCHAR(MAX) NOT NULL,
7     [SourceKey] NVARCHAR(255) NULL,
8
9     CONSTRAINT [PK_stg_ETL_Errors] PRIMARY KEY CLUSTERED ([ErrorID] ASC)
10 );
11

```

The Lookup component in Package 4 is configured to redirect non-matching rows to the error table rather than failing the entire package. A Row Count transform captures the count of unmatched rows in the variable User::NoMatchCount.

- FactPharmacySales table **before** task 6

Results		Messages														
nKey	MedicineKey	SupplierKey	CustomerKey	PaymentKey	TransactionID	Quantity	UnitPrice_LKR	DiscountRate	DiscountAmount_LKR	LineTotal_LKR	PrescriptionRequired	accm_txn_create_time	accm_txn_complete_time	txn_process_time_hours		
1	3382	49	1419	70	126	2	2023.86	0.0500	202.38	3845.33	0	2026-03-30 19:49:00.833	NULL	NULL		
2	3366	49	1421	70	110	1	3491.91	0.0600	209.51	3282.40	0	2026-03-30 19:49:00.833	NULL	NULL		
3	3287	51	1458	72	31	5	574.20	0.1000	287.10	2583.90	0	2026-03-30 19:49:00.833	NULL	NULL		
4	3368	50	1468	71	112	5	2910.52	0.1200	1746.31	12806.29	0	2026-03-30 19:49:00.833	NULL	NULL		
5	3373	51	1469	70	117	3	554.47	0.0500	83.17	1580.24	0	2026-03-30 19:49:00.833	NULL	NULL		
6	3388	52	1433	70	132	2	1475.02	0.0500	147.50	2802.54	0	2026-03-30 19:49:00.833	NULL	NULL		
7	3263	54	1434	71	7	3	84.34	0.1400	35.42	217.60	0	2026-03-30 19:49:00.833	NULL	NULL		
8	3376	51	1459	71	120	5	865.39	0.0100	43.26	4283.68	0	2026-03-30 19:49:00.833	NULL	NULL		
9	3392	50	1435	70	136	4	826.08	0.0800	264.34	3039.97	0	2026-03-30 19:49:00.833	NULL	NULL		
10	3364	50	1466	72	108	1	2011.96	0.0100	20.11	1991.84	0	2026-03-30 19:49:00.833	NULL	NULL		
11	3292	54	1453	72	36	2	1983.00	0.0400	158.64	3807.36	0	2026-03-30 19:49:00.833	NULL	NULL		
12	3361	51	1456	72	105	4	3412.23	0.1500	2047.33	11601.58	0	2026-03-30 19:49:00.833	NULL	NULL		
13	3293	49	1450	71	37	5	2889.43	0.0700	1011.30	13435.85	0	2026-03-30 19:49:00.833	NULL	NULL		
14	3318	54	1468	70	62	4	3116.09	0.1200	1495.72	10968.64	0	2026-03-30 19:49:00.833	NULL	NULL		
15	3267	56	1435	71	11	1	3125.06	0.0400	125.00	3000.06	0	2026-03-30 19:49:00.833	NULL	NULL		
16	3388	52	1469	71	132	2	1539.62	0.1100	338.71	2740.52	0	2026-03-30 19:49:00.833	NULL	NULL		
17	3295	52	1422	71	39	2	1924.36	0.0600	230.92	3617.80	0	2026-03-30 19:49:00.833	NULL	NULL		

- FactPharmacySales table **After** task 6

Results	Messages	SalesKey	DateKey	BranchKey	MedicineKey	SupplierKey	CustomerKey	PaymentKey	TransactionID	Quantity	UnitPrice_LKR	DiscountRate	DiscountAmount_LKR	LineTotal_LKR	PrescriptionRequired	accm_txn_create_time	accm_txn_complete_time
1		1	20230417	161	3382	49	1419	70	126	2	2023.86	0.0500	202.38	3845.33	0	2026-03-30 18:26:02.767	2026-03-28 10:42:08.000
2		2	20220606	157	3366	49	1421	70	110	1	3491.91	0.0600	209.51	3282.40	0	2026-03-30 18:26:02.767	2026-03-30 23:42:08.000
3		3	20230410	159	3287	51	1458	72	31	5	574.20	0.1000	287.10	2583.90	0	2026-03-30 18:26:02.767	2026-03-29 11:42:08.000
4		4	20221030	159	3368	50	1468	71	112	5	2910.52	0.1200	1746.31	12806.29	0	2026-03-30 18:26:02.767	2026-03-27 05:42:08.000
5		5	20220212	158	3373	51	1469	70	117	3	554.47	0.0500	83.17	1580.24	0	2026-03-30 18:26:02.767	2026-03-30 09:42:08.000
6		6	20220103	157	3388	52	1433	70	132	2	1475.02	0.0500	147.50	2802.54	0	2026-03-30 18:26:02.767	2026-03-31 09:42:08.000
7		7	20220930	159	3263	54	1434	71	7	3	84.34	0.1400	35.42	217.60	0	2026-03-30 18:26:02.767	2026-03-30 09:42:08.000
8		8	20220622	155	3376	51	1459	71	120	5	865.39	0.0100	43.26	4283.68	0	2026-03-30 18:26:02.767	2026-03-29 15:42:08.000
9		9	20230628	159	3392	50	1435	70	136	4	826.08	0.0800	264.34	3039.97	0	2026-03-30 18:26:02.767	2026-03-28 05:42:08.000
10		10	20230330	160	3364	50	1466	72	108	1	2011.96	0.0100	20.11	1991.84	0	2026-03-30 18:26:02.767	2026-03-31 10:42:08.000
11		11	20230401	157	3292	54	1453	72	36	2	1983.00	0.0400	158.64	3807.36	0	2026-03-30 18:26:02.767	2026-03-26 19:42:08.000
12		12	20230416	156	3361	51	1456	72	105	4	3412.23	0.1500	2047.33	11601.58	0	2026-03-30 18:26:02.767	2026-03-30 02:42:08.000
13		13	20230120	157	3293	49	1450	71	37	5	2889.43	0.0700	1011.30	13435.85	0	2026-03-30 18:26:02.767	2026-03-31 10:42:08.000
14		14	20220723	161	3318	54	1468	70	62	4	3116.09	0.1200	1495.72	10968.64	0	2026-03-30 18:26:02.767	2026-03-29 18:42:08.000
15		15	20220522	159	3267	56	1435	71	11	1	3125.06	0.0400	125.00	3000.06	0	2026-03-30 18:26:02.767	2026-03-30 19:42:08.000
16		16	20220318	157	3388	52	1469	71	132	2	1539.62	0.1100	338.71	2740.52	0	2026-03-30 18:26:02.767	2026-03-30 15:42:08.000
17		17	20230322	160	3295	52	1422	71	39	2	1924.36	0.0600	230.92	3617.80	0	2026-03-30 18:26:02.767	2026-03-30 07:42:08.000

6.5 Validation

After running Package 4, the following validation queries confirm the accumulating fact update was successful

SQLQuery7.... GF66 (56))* ✕ SQLQuery6.sq...A GF66 (81)) SQLQuery94.s...A GF66 (83))

```
3  -- Check updated rows
4  SELECT TransactionID, accm_txn_create_time,
5         accm_txn_complete_time, txn_process_time_ho
6  FROM FactPharmacySales
7  WHERE accm_txn_complete_time IS NOT NULL
8  ORDER BY TransactionID;
9
10 -- Summary statistics
11 SELECT COUNT(*) AS UpdatedRows,
12        AVG(txn_process_time_hours) AS AvgHours,
13        MIN(txn_process_time_hours) AS MinHours,
14        MAX(txn_process_time_hours) AS MaxHours
15 FROM FactPharmacySales
16 WHERE accm_txn_complete_time IS NOT NULL;
17
18
```

119 % No issues found

Results Messages

	TransactionID	accm_txn_create_time	accm_txn_complete_time	txn_process_time_hours	
1	1	2026-03-30 19:49:00.833	2026-03-28 02:42:08.000	-65.00	
2	1	2026-03-30 19:49:00.833	2026-03-27 06:42:08.000	-85.00	
3	1	2026-03-30 19:49:00.833	2026-03-31 14:42:08.000	19.00	
4	1	2026-03-30 19:49:00.833	2026-03-29 16:42:08.000	-27.00	
5	1	2026-03-30 19:49:00.833	2026-03-30 05:42:08.000	-14.00	
6	1	2026-03-30 19:49:00.833	2026-03-27 08:42:08.000	-83.00	
7	1	2026-03-30 19:49:00.833	2026-03-30 07:42:08.000	-12.00	
8	1	2026-03-30 19:49:00.833	2026-03-30 23:42:08.000	4.00	
9	1	2026-03-30 19:49:00.833	2026-03-29 01:42:08.000	-42.00	
10	1	2026-03-30 19:49:00.833	2026-03-28 10:42:08.000	-57.00	
11	1	2026-03-30 19:49:00.833	2026-03-27 05:42:08.000	-86.00	
12	1	2026-03-30 19:49:00.833	2026-03-28 23:42:08.000	-44.00	
	UpdatedRows	AvgHours	MinHours	MaxHours	
1	50000	-36.718140	-96.00	23.00	